Shikhar Saxena

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Education

Thapar Institute of Engineering and Technology, Patiala Aug 2023 – Jun 2027 (Expected)
- B.E. Computer Science and Engineering | CGPA: 8.59/10

Army Public School, Bareilly Class of 2023
- **12th:** 92.4%, **10th:** 93%

Experience

TruckHai, Backend Intern Jan 2026 – Apr 2026
- Built Go (Gin, GORM) **REST microservices** for fleet dispatch, modelling the trip lifecycle as a **finite-state machine** with **ACID transactions** so that invalid state transitions are rejected and concurrent updates stay consistent.
- Removed race conditions and row-lock contention in **PostgreSQL** through **composite indexing** and query-plan tuning, and added **RBAC middleware** with audit-logged admin impersonation across **5+ user roles**.

Voxomos Systems Pvt Ltd, Software Intern Jun 2026 – Jul 2026
- Built and benchmarked CPU and GPU (**CUDA**, **AMD HIP**) implementations of vector similarity search, profiling bottlenecks to compare performance across hardware.
- Load-tested LLM serving frameworks (**vLLM**, **SGLang**, **TensorRT-LLM**), holding **14–15 ms/token** latency and reducing first-response latency spikes by tuning request concurrency.

Projects

Relay |*Python, FastAPI, MCP, LangGraph, PostgreSQL, Redis, Kubernetes, Helm, Next.js* **github.com/Shikhar1808/Relay**
- Built a FastAPI **MCP gateway** that serves **11 tools from 3 custom MCP servers** through one Streamable HTTP endpoint, with per-user API keys, role-based tool permissions and a **120 calls/min token-bucket rate limit**; added a **LangGraph** agent that acts as the signed-in user and confirms before any write.
- Isolated server failures with per-server timeouts and **30 s health probes**, and monitored **p95 latency**, errors and uptime in **Prometheus/Grafana**.
- Deployed on **Kubernetes (k3s)** with **Helm** and per-tenant namespaces behind **deny-by-default NetworkPolicies**; secured with **AES-256-GCM**-encrypted credentials and an **HMAC-chained audit log**.

CivicShield — Emergency Response & FIR System | *MERN, Solidity, LangChain* **github.com/Shikhar1808/HackMol-6.0**
- Built a tamper-proof FIR system with a **Solidity** smart contract tracking complaint states, **IPFS** evidence storage and an SLA job that auto-escalates unattended complaints.
- Classified complaints into **10 categories** and **3 priority levels** with **LangChainJS + GPT-4**, behind a **React/TypeScript** dashboard and **Express/MongoDB** backend.

Zenith Website | *NodeJS, ExpressJS, MongoDB, JWT Auth, NGINX, AWS* **github.com/Shikhar1808/ZenithCryptic**
- Built a competition platform serving **1000+ concurrent users** at **sub-100 ms** response times; designed real-time leaderboard calculation and stateless **JWT** auth to remove database bottlenecks during submission spikes.
- Secured the deployment with **Cloudflare** DDoS mitigation and **NGINX** reverse-proxy rate limiting and load balancing.

Technologies and Skills

Languages: C++, Go, Python, JavaScript, SQL
Backend & Web: Microservices, REST APIs, Node.js, Express.js, FastAPI, Gin, GORM, React.js, Tailwind CSS, Socket.io, JWT, RBAC
Databases & DevOps: PostgreSQL, MongoDB, MySQL, Redis, Docker, Linux, GitHub Actions (CI/CD), NGINX, Prometheus, Grafana, Git, AWS, Kubernetes, Helm
AI & GPU Computing: PyTorch, CUDA, vLLM, LangChain, LangGraph, MCP
Core CS: Data Structures and Algorithms, OOPs, DBMS, Operating Systems, Computer Networks

Leadership and Achievements

Tech Lead, Thapar Institute Counselling Cell Oct 2025 – Present
SPHNIX'24 Hackathon – 1st Runner Up, MNIT Jaipur Nov 2024
NVIDIA Certified: Accelerated Computing (Modern CUDA C++, CUDA Python), AI on Jetson Nano, Generative AI & Deep Learning 2025 – 2026
LeetCode: 417+ problems solved, 1551 Contest Rating (DSA: arrays, graphs, DP, trees, stacks, queues, heap)